

RISHABH GOEL

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Curious and driven technologist who blends math, code, and creativity to explore complex ideas. Passionate about scientific discovery, mentoring peers, and making hard concepts approachable.

EDUCATION

Rock Ridge High School, Ashburn VA - High School Diploma | Sep 2020 - Jun 2025

Relevant coursework: AP Physics C: Mechanics, AP Calculus BC, AP Statistics, AP Computer Science A

University of Maryland, College Park MD - Physics Major | Sep 2025 - May 2029

Relevant coursework: Introductory Physics: Fields; Introductory Physics: Oscillations and Waves; Experimental Physics I: Mechanics and Waves; Experimental Physics II: Electricity and Magnetism; Multivariable Calculus, Linear Algebra, Differential Equations I and II (Honors); Special Problems in Physics; Object Oriented Programming I and II (exemption exam)

EXPERIENCE & PROJECTS

ZeroClone - Open-Source Creator | Apr 2025 - Present

- Building a general learning algorithm for two-player perfect-information games, inspired by Google DeepMind's AlphaZero.
- Implemented batched-GPU self-play pipeline and plug-and-play game adapters in C++ and Python.

ResearchTree - Open-Source Creator | May 2026 - Present

- Built a framework for exploring AI research ideas through structured search and experimentation.
- Added Dockerized workflows for evaluating candidate branches with repeatable scorers and preserved artifacts.

UMD LHCb Group - Undergraduate Researcher | Dec 2025 - Present

- Using bootstrapping methods to estimate uncertainty in one of the trigger mechanisms in the CERN LHC.
- Collaborated with graduate students and worked with large datasets using ROOT and similar tools.

Rock Ridge HS Competitive Programming Club - President | Sep 2021 - Jun 2025

- Led and coached 15 members; placed Top-10 in multiple Chesapeake-region university contests.
- Organized weekly problem-solving workshops and collectively solved 30+ 1800-rated problems on Codeforces.

Yogic Transformers FTC Robotics Team - Autonomous Algorithms Lead | Sep 2021 - Jun 2023

- Developed sensor-fusion and vision pipelines for fully autonomous navigation; qualified for state semi-finals twice.

Quantiphi - Machine Learning Engineer Intern | Jun 2020 - Jul 2020

- Prototyped a computer-vision research project under senior mentorship and presented findings to leadership.

SKILLS

Languages: Python; C++; Java; Bash/Zsh; Embedded C/C++ (Arduino)

AI / ML & Data: PyTorch; Reinforcement Learning (MCTS, self-play); NumPy/Pandas

Systems & DevOps: Linux admin; Docker; CMake/build pipelines; Git/GitHub

Math & Science: Mechanics; Electromagnetism; Probability/Statistics; Calculus

Agentic Coding: Claude Code; Codex

INTERESTS

Particle physics, cosmology, deep learning, character-driven literature ("The Goldfinch," "A Little Life," "Pachinko") and cinema ("Whiplash," "The Pianist," "The Shawshank Redemption").